Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Accuracy as a Fraction

> Prepare Accuracy as a Fraction: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain why overall **accuracy** can hide failure on a less common class

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/12, ai-4/16. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Training and Testing

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a stated task

Reference example, not the only acceptable wording: Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Look for: Design revealing tests for a function

Reference example, not the only acceptable wording: 4, 0 and 6.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

Concept to explain

Accuracy is correct **prediction**s divided by all **prediction**s.

Term	Meaning
accuracy	The number of correct predictions divided by the total number of scored predictions.
numerator	The top number of a fraction, identifying the counted portion.
denominator	The bottom number of a fraction, identifying the number of equal parts or total cases considered.

Term	Meaning
percentage	A proportion expressed per hundred; multiply a fraction by 100 to express it as a percent.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer **count**ers or a calculator after learners show the setup. Fictional records are supplied; personal data and paid ac**count**s are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Begin with a majority guess that looks successful under overall **accuracy**, then keep both class **denominator**s visible.

Introduce **balanced accuracy** as an equal average of two correct fractions, without treating it as a universal replacement for task-specific evaluation.

Readiness answer

They are in the blocked-reference row and clear-**prediction** column: the lower-left cell. These are missed blocked cases.

Printed lesson and workbook

Twenty **count**ers with sixteen labelled clear and four blocked

Blank confusion matrices and fraction strips

Optional Python 3 editor with no packages required

Apply the connected idea: Summaries with counts and proportions

Teach Week 16 as a two-session investigation using fictional data. A fictional poll asks whether a short activity was useful. Responses are Yes, No or Unknown.

We will report Yes among known responses and also disclose unknown responses. Choosing a **denominator** changes the question being answered.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

3/12, which is 1/4.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can compare **accuracy** with class-specific performance in an imbalanced evaluation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Accuracy as a Fraction using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Model A: 7 correct out of 10 → 7/10 = **70%**.

2. Model B: 18 correct out of 20 → 18/20 = 90%.

3. Model B has higher measured **accuracy** on these tests.

4. Still ask: were the test cases fair and were some errors more costly?

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Nine of twelve held-out classifications are correct.

The teaching case

Compute **accuracy** and state the **denominator**.

Exact explanation to model

> ### Why the result follows
>
> 9/12=3/4=75%. **Count** all scored cases, including errors.

Explain the added worked example

Explain the added worked example · 1

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

Explain the added worked example · 2

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Explain the added worked example · 3

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

Explain the added worked example · 4

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first **percentage** is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

Explain the added worked example · 5

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer **count**s in the stated order.

It stops when either reference class is absent. Then report the observed **count**s and the unavailable class result; do not invent a **denominator** or silently substitute zero.

Key terms

Term	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

Apply the connected idea: Summaries with counts and proportions

A fictional poll asks whether a short activity was useful. Responses are Yes, No or Unknown. We will report Yes among known responses and also disclose unknown responses.

Choosing a **denominator** changes the question being answered.

The printed **proportion** is 0.75, or 75 percent, among four known responses. Yes among all five supplied responses is 0.6, or 60 percent. Both calculations are correct when labelled; they describe different **denominator**s.

Unknown must not silently become No.

1/2 = 50 percent among known; 1/4 = 25 percent among all supplied. Exit: No. The **denominator** is known volunteers, and participation may not represent all students.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Model the reasoning aloud · 6

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Accuracy as a Fraction

A fictional shelf classifier predicts repair needed or not needed.

The facts we will use

Supplied facts

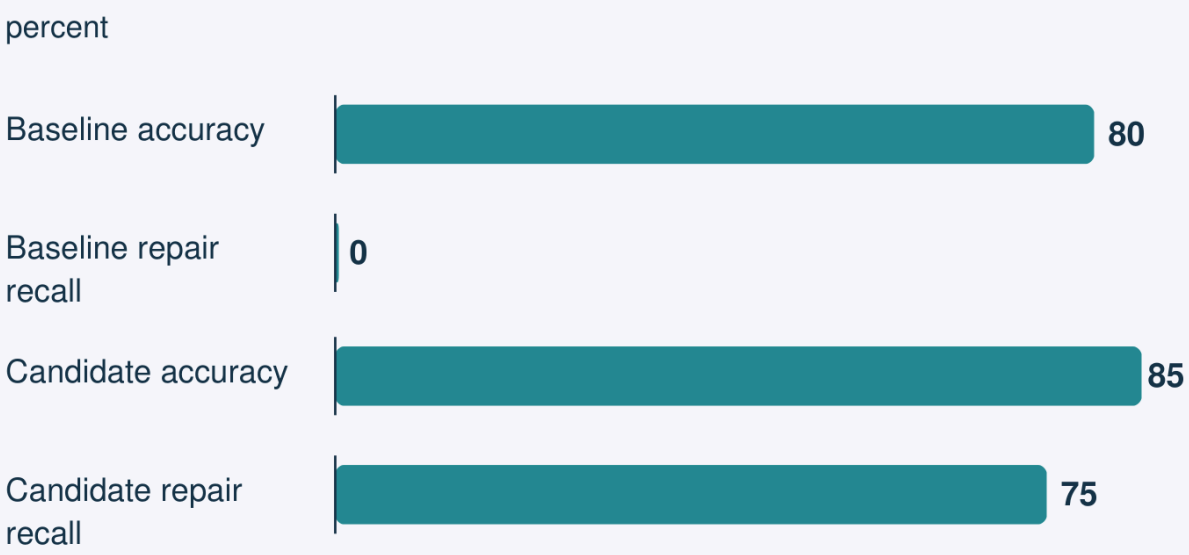
Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Supplied facts

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.

AI ACADEMY / AI-2/12

Accuracy can hide missed repair cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reference population

20 labelled cases, including four repair-needed books.



2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.



3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.



4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reference population	20 labelled cases, including four repair-needed books.
2. Baseline	16/20=80% accuracy; catches 0/4 needed cases.
3. Candidate	(3+14)/20=85% accuracy; catches 3/4=75% needed cases.
4. Compare costs	Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

What this example does not establish

Keep the stated limit

These selected cases do not establish future rates or specify the real cost of each error.

Change one thing in the pictured case

Change one thing

Change: A candidate instead marks every book repair needed.

Predicted result

Prediction: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

Why the result follows

Reason: All 16 not-needed books become false alarms.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Accuracy as a Fraction, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Nine of twelve held-out classifications are correct.

2. State the decision to explain

Write the question to explain: Compute **accuracy** and state the **denominator**.

3. Reveal the worked reasoning

Reveal after **prediction**s: 9/12=3/4=75%. **Count** all scored cases, including errors.

Hold back the independent task

Keep for an independent attempt: A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: baseline has FP=0,FN=4, finding none of four blocked cases. Candidate has FP=4,FN=0, finding all four but adding four false alarms. Both have sixteen correct **prediction**s.

Equal averaging gives baseline (100+0)/2=50 percent and candidate (75+100)/2=87.5 percent. Exit: request the full labelled matrix, class **count**s, sampling method, **training**-derived baseline and task consequences.

The worked always-clear rule misses all four blocked cases.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve false negatives and reference-row totals from Week 13.
7 to 18 minutes	Use counters to score the always-clear baseline and complete W1.
18 to 32 minutes	Compare the candidate matrix. Calculate ordinary and class-specific correct fractions in W2.
32 to 45 minutes	Trace the code and read the visual. Answer the guided question and explain equal class weighting.
Session 2 start to 8 minutes	Retrieve each denominator and why a high overall score can hide rare-class failure.

Stage	Teacher action
8 to 20 minutes	Complete W3 with a changed class mix. Separate the arithmetic assumption from claims about deployment.
20 to 35 minutes	Collect W4 independently before support. Require fractions before rounded percentages and the error tradeoff.
35 to 45 minutes	Use W5 and the exit question to discuss missing classes. Offer the fresh retry after feedback and record support.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Reasoning to listen for

Teacher check: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Summaries with counts and proportions

Target: **Count** categories with a stated **denominator**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

Reasoning to listen for

Teacher check: Both are 50 percent. **Count**s and sample sizes differ: 2/4 versus 3/6.

Individual check: For Summaries with **count**s and **proportion**s, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Why is 100% repair detection not sufficient to declare the changed classifier best?

Reasoned teacher answer

It catches all four repairs but wrongly flags all 16 others, creating many extra checks; consequences must be assessed.

Changed-case reasoning: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Accuracy as a Fraction investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Score two **prediction** sheets, simplify or convert each fraction, compare **percentage**s, then inspect which cases were wrong.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

High **accuracy** means both classes work well: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Balanced accuracy fixes the model: It changes the summary weighting, not any **prediction** or underlying error.

An absent class has zero measured performance: Its fraction has a zero **denominator** and is unavailable. Report missing coverage rather than inventing a value.

Check these misconceptions

A successful run proves the conclusion: A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: No, Yes, Unknown, No, Unknown gives 1 Yes of 3 known, about 33.3 percent, with 2 unknown.

Accept 1/3 without rounding.

A convenient shortcut preserves meaning: Dividing by known when it equals 0 causes ZeroDivisionError. Test known > 0 before division and report no known responses if none exist.

Reporting 0 percent would wrongly suggest known responses with no Yes answers.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Accuracy as a Fraction questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

9/12=3/4=75%. **Count** all scored cases, including errors.

Independent question

Independent question · withhold the key

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

Expected independent answer

Teacher key · Expected independent answer

9/12 is 75%; 8/10 is **80%**. The second fraction is higher, although differing test sets can make a direct model comparison unfair.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not **count** a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “More correct answers always means higher **accuracy**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

A=**80%**, B=**60%**; combined 26/40=65%. The unweighted average is **70%**, which gives the ten-case class the same weight as the thirty-case class. State whether the question concerns a typical class or a typical tested case.

Explanations for the additional workbook examples

W1 Score the majority baseline

Workbook question · W1

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

Teacher reasoning · W1

The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

W2 Compare both class fractions

Workbook question · W2

For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Teacher reasoning · W2

Accuracy 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

W3 Explain a changed class mix

Workbook question · W3

For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

Teacher reasoning · W3

Baseline ordinary 8/16=50 percent and balanced (100+0)/2=50 percent. Candidate ordinary 14/16=87.5 percent and balanced (75+100)/2=87.5 percent. Within-class fractions stayed fixed while class **count**s changed.

This arithmetic assumption does not prove unchanged real-world behavior on a new population.

W4 Assess a new comparison independently

Workbook question · W4

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Teacher reasoning · W4

Baseline **accuracy** 18/20=90 percent; class fractions 100 percent and 0 percent; **balanced accuracy** 50 percent.

Candidate **accuracy** 16/20=80 percent; clear fraction 15/18=5/6 about 83.3 percent, blocked fraction 1/2=50 percent; **balanced accuracy** (5/6+1/2)/2=2/3 about 66.7 percent. Candidate FP=3,FN=1.

It finds one blocked case but adds three false alarms; task consequences and the small sample matter.

W5 Limit claims from missing classes

Workbook question · W5

An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

Teacher reasoning · W5

Ordinary **accuracy** is 10/10=100 percent on the observed clear cases. The blocked-class fraction has **denominator** zero, so it cannot be measured.

Our two-class summary function stops with ValueError; report the missing coverage instead of claiming perfect performance on both classes.

Summaries with counts and proportions — W1 State the denominator

In the worked poll, why is the known **denominator** 4 rather than 5?

Only Yes and No are included; one Unknown is excluded and must be disclosed.

Summaries with counts and proportions — W2 Compare proportions

Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

Both are 50 percent. **Count**s and sample sizes differ: 2/4 versus 3/6.

Summaries with counts and proportions — W3 Repair zero division

All three responses are Unknown. What should the program report?

No known responses, with 3 unknown. Do not divide by zero or report 0 percent as a measured **proportion**.

Summaries with counts and proportions — W4 Summarize independently

Process Yes, Yes, No, Unknown, No, Yes. Report Yes **count**, known **count**, unknown **count** and Yes **proportion** among known.

3 Yes, 5 known, 1 unknown; 3/5 = 0.6 = 60 percent.

Summaries with counts and proportions — W5 Avoid overclaiming

> Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

No. The **denominator** is known volunteers, and participation may not represent all students.

Assess independent transfer

> #### Independent question · withhold the key
>
> Selected case: New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.
>
> The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.
>
> Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Selected case answer

> #### Teacher key · Selected case answer
>
> Baseline **accuracy** 18/20=90 percent; class fractions 100 percent and 0 percent; **balanced accuracy** 50 percent.
>
> Candidate **accuracy** 16/20=80 percent; clear fraction 15/18=5/6 about 83.3 percent, blocked fraction 1/2=50 percent; **balanced accuracy** (5/6+1/2)/2=2/3 about 66.7 percent. Candidate FP=3,FN=1.
>
> It finds one blocked case but adds three false alarms; task consequences and the small sample matter.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Accuracy as a Fraction to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Report correct recommendations / all test users, alongside the raw test **count**.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with **count**ers and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 13: Types of Classification Error. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use **count**ers and write each fraction with its reference-class total before converting to a **percentage**. Supply blank **denominator** boxes rather than answers.

Extension: express ordinary **accuracy** as the class-size-weighted average of the two correct fractions; for the candidate use (16/20)(12/16)+(4/20)(4/4)=16/20.

Require W4 baseline ordinary 90 percent and balanced 50 percent; candidate ordinary 80 percent and balanced 2/3 about 66.7 percent, with FP=3,FN=1.

Fresh retry: evaluation 12 clear and 4 blocked, baseline [[12,0],[4,0]], candidate [[9,3],[1,3]], **training** majority clear. Baseline ordinary 75 percent, class fractions 100 and 0 percent, balanced 50 percent.

Candidate ordinary 12/16=75 percent, both class fractions 75 percent, balanced 75 percent, FP=3,FN=1. Require both **denominator**s and no automatic-winner claim.

Week 15 covers precision recall and **denominator**s. Carry forward the matrix and the question answered by each fraction.

Teacher reference: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.balanced_**accuracy**_score.html (equal average of class recalls; reviewed 2026-09-28).

Readiness and reassessment for Summaries with counts and proportions

Use fraction bars with separate unknown cards. Extension: show both **denominator**s side by side with explicit labels.

Independent W4 **evidence**: 3 Yes, 5 known, 1 unknown; 3/5 = 0.6 = 60 percent. Record whether the learner needed no help, a hint or explanation.

Fresh retry: No, Yes, Unknown, No, Unknown gives 1 Yes of 3 known, about 33.3 percent, with 2 unknown. Accept 1/3 without rounding.

Plan the delayed return

Suggested return points in this level: ai-2/13, ai-2/15, ai-2/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Accuracy as a Fraction lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain why overall **accuracy** can hide failure on a less common class

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI ac**count** is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives.

Compare with the baseline.

Supported practice answer

Reasoned practice answer: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Check every learner before discussion: Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: A candidate instead marks every book repair needed. Require a **prediction** and a reason before comparing. Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

Reason: All 16 not-needed books become false alarms.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

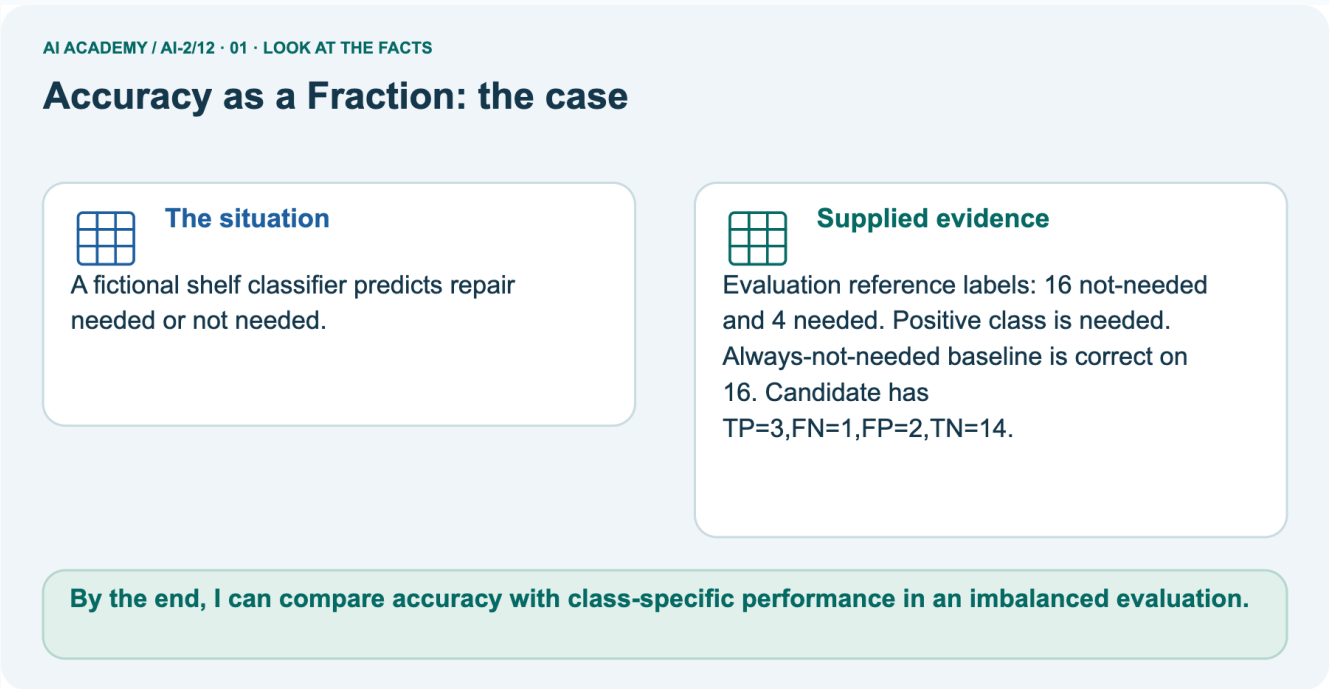
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Accuracy as a Fraction graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Accuracy as a Fraction: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

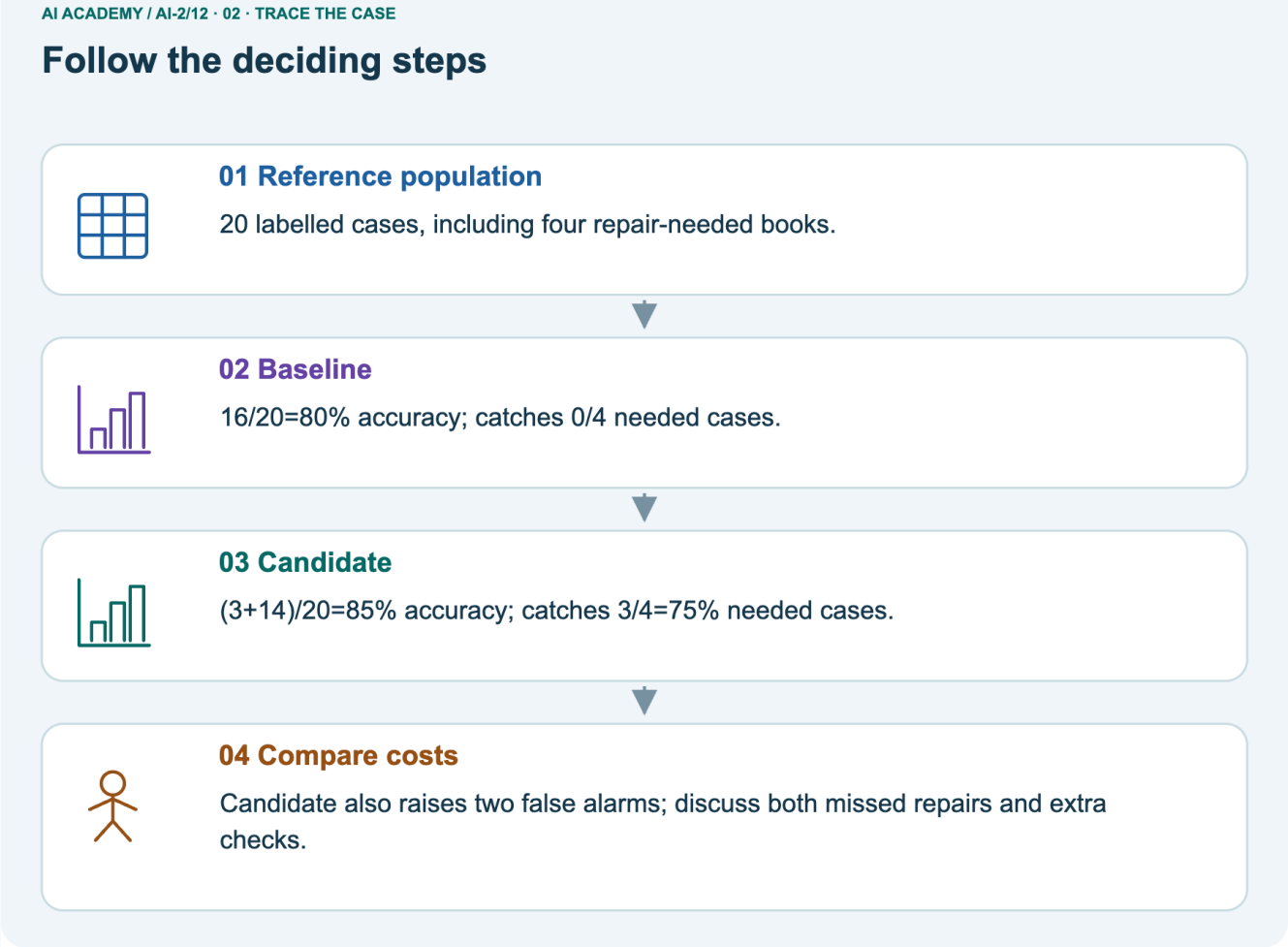
Candidate has TP=3,FN=1,FP=2,TN=14.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

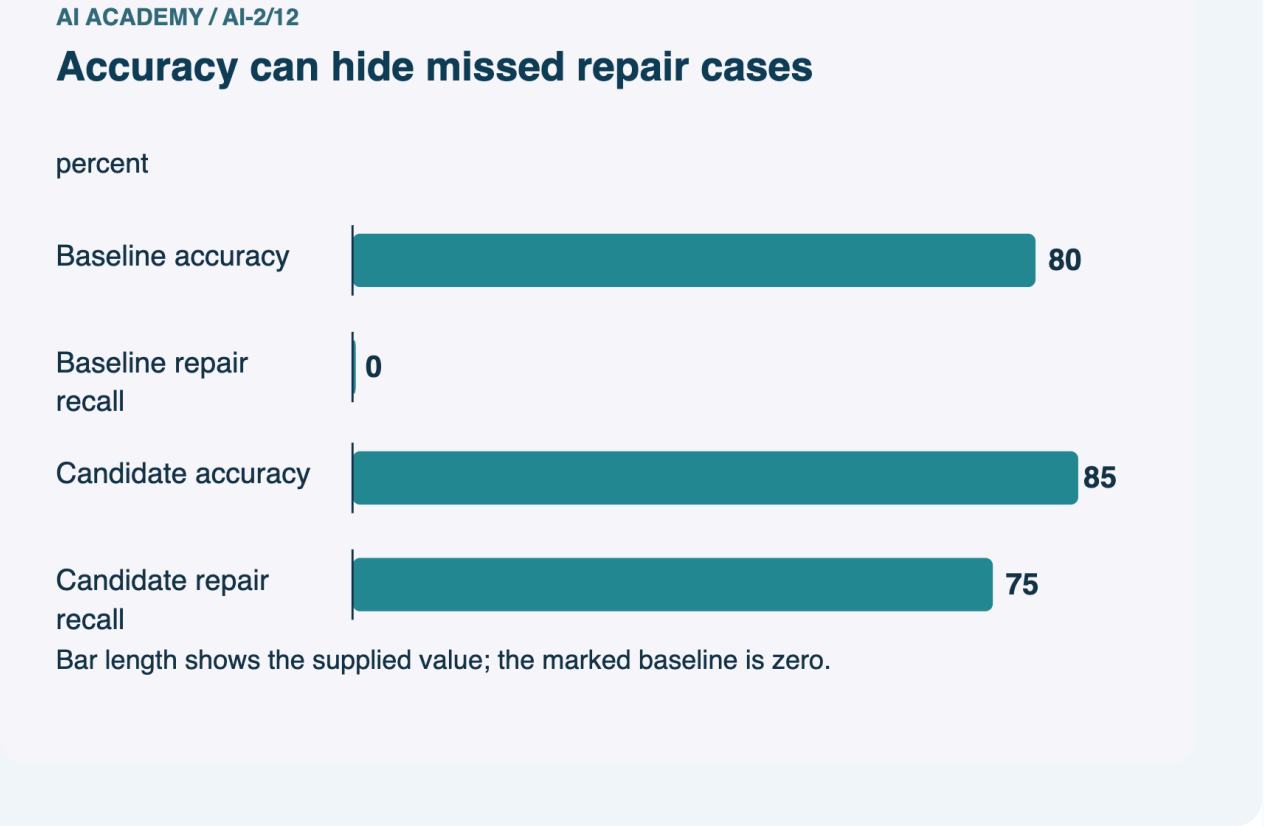
Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Accuracy can hide missed repair cases



Follow the labelled picture with the matching explanation. Open the vector diagram

Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

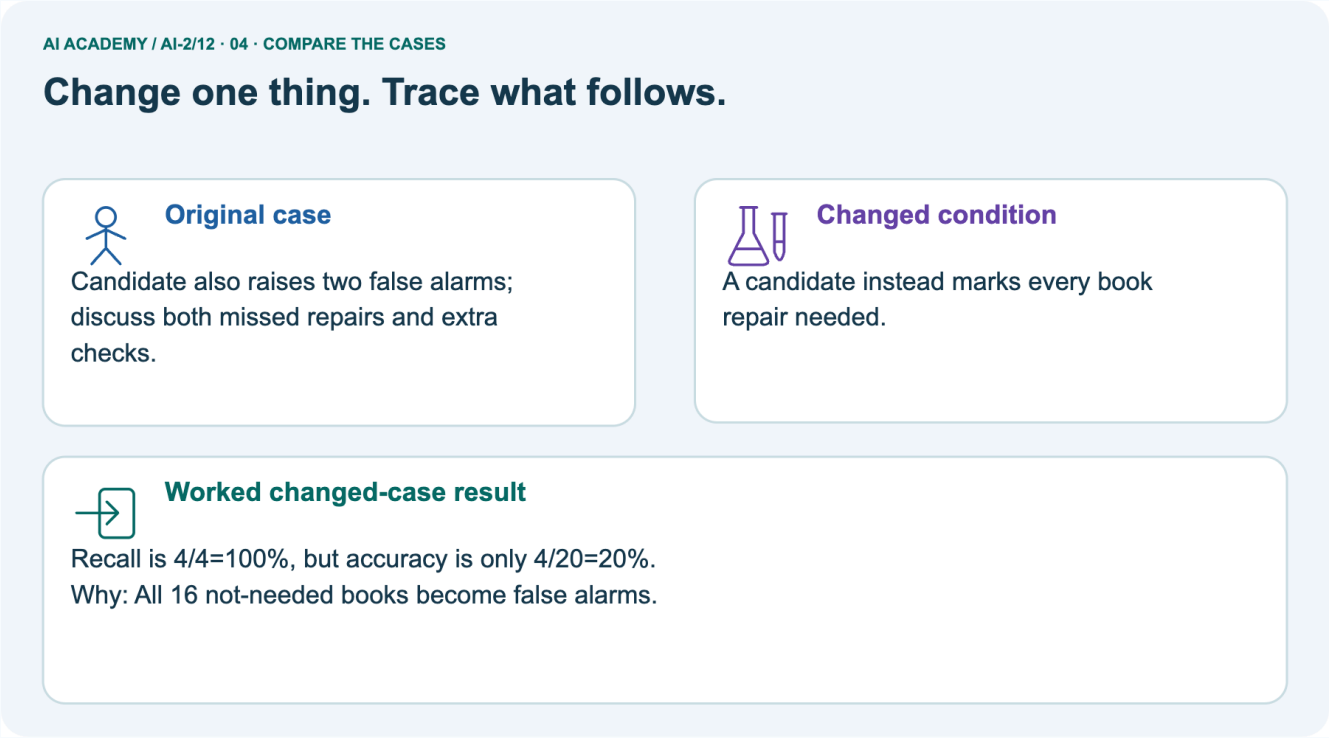
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

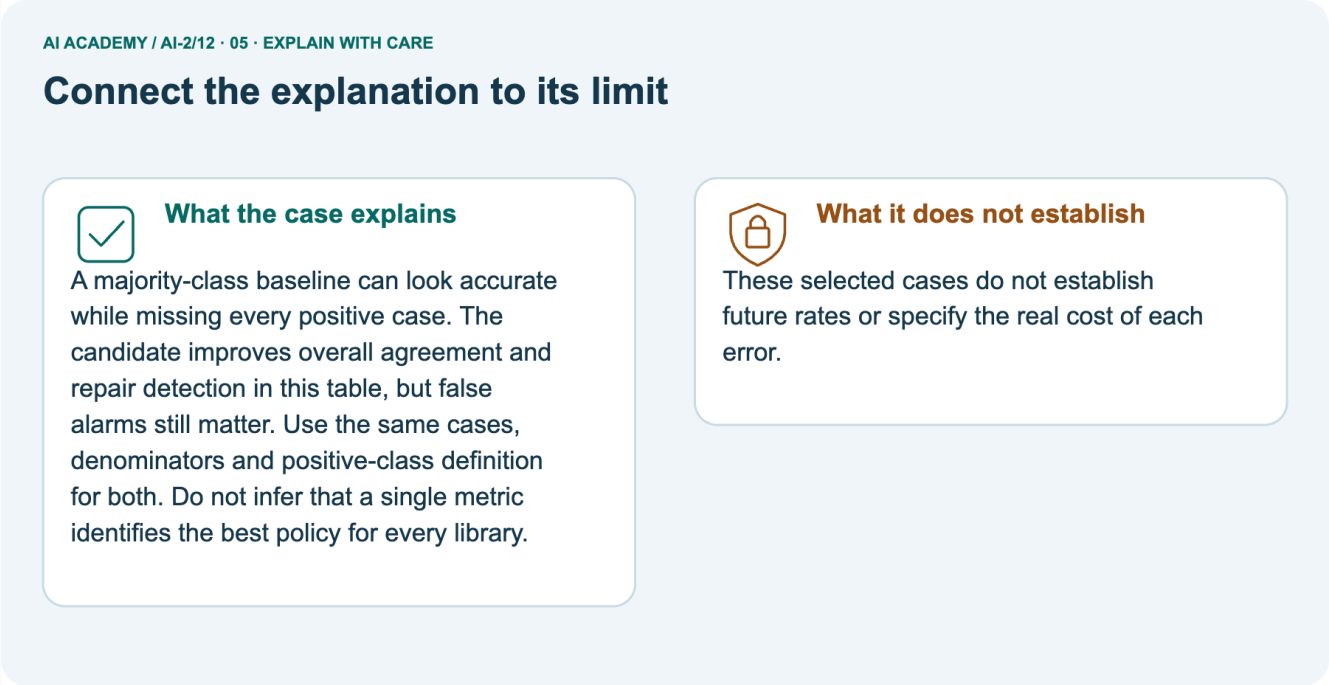
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: All 16 not-needed books become false alarms.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Limit: These selected cases do not establish future rates or specify the real cost of each error.

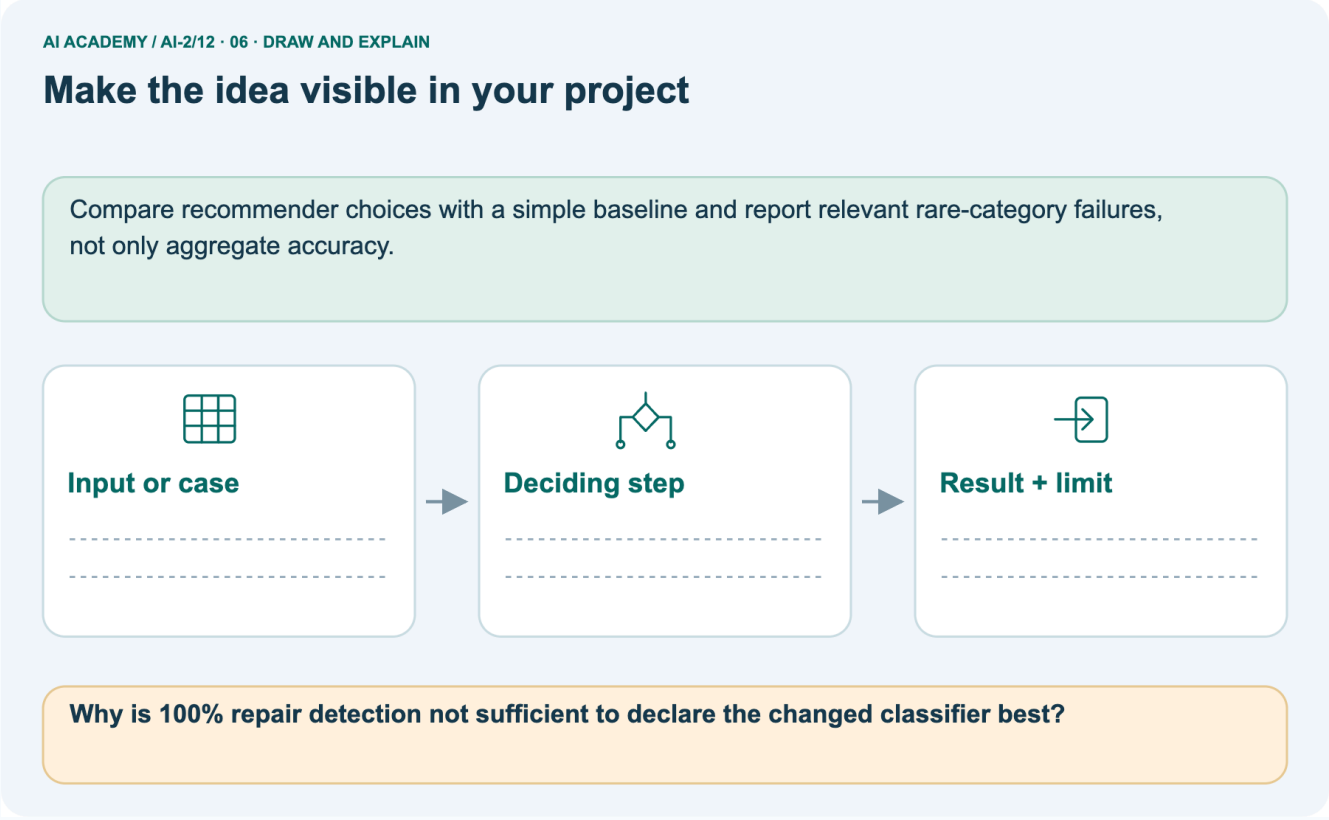
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These selected cases do not establish future rates or specify the real cost of each error.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It catches all four repairs but wrongly flags all 16 others, creating many extra checks; consequences must be assessed.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.